```
postgres-# ORDER BY s.last_name, c.course_code;
 student_name  | course_code |         course_name          | grade | enrollment_date
---------------+-------------+------------------------------+-------+-----------------
 Alice Johnson | CS101       | Introduction to Programming | A     | 2026-10-19
 Alice Johnson | CS201       | Data Structures             | B+    | 2026-10-19
 Bob Smith     | CS101       | Introduction to Programming | A-    | 2026-10-19
 Bob Smith     | MATH101     | Calculus I                  | B     | 2026-10-19
 Carol Wilson  | CS201       | Data Structures             | A     | 2026-10-19
 Carol Wilson  | MATH101     | Calculus I                  | A-    | 2026-10-19
(6 rows)


postgres=#
postgres=# -- All students in a course
postgres=# SELECT s.first_name, s.last_name, se.grade
postgres-# FROM students s
postgres-# JOIN student_enrollments se ON s.student_id = se.student_id
postgres-# JOIN courses c ON se.course_id = c.course_id
postgres-# WHERE c.course_code = 'CS101';
 first_name | last_name | grade
------------+-----------+-------
 Alice      | Johnson   | A
 Bob        | Smith     | A-
(2 rows)


postgres=#
postgres=# -- All courses for a student
postgres=# SELECT c.course_code, c.course_name, c.credits, se.grade
postgres-# FROM courses c
postgres-# JOIN student_enrollments se ON c.course_id = se.course_id
postgres-# JOIN students s ON se.student_id = s.student_id
postgres-# WHERE s.email = 'alice@university.edu';
```

```
postgres=#
postgres=# -- All courses for a student
postgres=# SELECT c.course_code, c.course_name, c.credits, se.grade
postgres-# FROM courses c
postgres-# JOIN student_enrollments se ON c.course_id = se.course_id
postgres-# JOIN students s ON se.student_id = s.student_id
postgres-# WHERE s.email = 'alice@university.edu';
 course_code |         course_name         | credits | grade
-------------+-----------------------------+---------+-------
 CS101       | Introduction to Programming |       3 | A
 CS201       | Data Structures             |       4 | B+
(2 rows)


postgres=# \dt lab8.*
              List of relations
 Schema |        Name         | Type  |  Owner
--------+---------------------+-------+----------
 lab8   | blog_posts          | table | postgres
 lab8   | comments            | table | postgres
 lab8   | courses             | table | postgres
 lab8   | departments         | table | postgres
 lab8   | employees           | table | postgres
 lab8   | student_enrollments | table | postgres
 lab8   | students            | table | postgres
 lab8   | user_profiles       | table | postgres
 lab8   | users               | table | postgres
(9 rows)


postgres=#
postgres=# SELECT c.relname AS table_name, con.conname AS foreign_key,
postgres-#        pg_get_constraintdef(con.oid) AS definition
postgres-# FROM pg_constraint con
postgres-# JOIN pg_class c ON c.oid = con.conrelid
postgres-# JOIN pg_namespace n ON n.oid = c.relnamespace
postgres-# WHERE n.nspname = 'lab8' AND con.contype = 'f'
postgres-# ORDER BY c.relname;
     table_name      |            foreign_key             |                                 definition
---------------------+------------------------------------+-----------------------------------------------------------------------------
 comments            | comments_post_id_fkey              | FOREIGN KEY (post_id) REFERENCES blog_posts(post_id) ON DELETE CASCADE
 employees           | employees_dept_id_fkey             | FOREIGN KEY (dept_id) REFERENCES departments(dept_id) ON DELETE RESTRICT
 student_enrollments | student_enrollments_student_id_fkey | FOREIGN KEY (student_id) REFERENCES students(student_id) ON DELETE CASCADE
 student_enrollments | student_enrollments_course_id_fkey  | FOREIGN KEY (course_id) REFERENCES courses(course_id) ON DELETE CASCADE
 user_profiles       | user_profiles_user_id_fkey         | FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE
(5 rows)


postgres=#
```